Matrices

Application of Matrices:

1. Matrices, being the organization of data into columns and rows, can have many applications in representing demographic data, in computer and scientific applications, among others. They can be used as a representation of data or as a tool to solve specific problems.
2. Matrices are used to represent real-world data about specific populations, such as the number of people who have a specific trait. They can also be used to model projections in population growth.
3. In the field of computing, matrices are used in message encryption. They are used to create three-dimensional graphic images and realistic looking motion on a two-dimensional computer screen and also in the calculation of algorithms that create Google page rankings.
4. Matrices are used to compress electronic information and play a role in storing fingerprint information. Errors in electronic transmissions are identified and corrected with the use of matrices.
5. The discipline of physics also uses matrices to calculate battery power outputs and resistor conversion of electrical energy into a more efficient form. Matrices are used to solve problems involving Kirchoff's laws of voltage and current.
6. Computer science also relies heavily on matrices. Tasks such as projecting a three-dimensional image onto a two-dimensional screen and encrypting message codes are two areas in which matrices are used. Matrices and their inverses are necessary for programmers to code encrypted messages, and these codes are used to run Internet programs, such as search systems.

Matrix: Definition: A system of any mn numbers arranged in a rectangular array of m rows and n columns is called matrix of order $m \times n$ or m by n is defined as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}_{m\times n}$$

Where the symbols a_{ij} represent any numbers. The elements a_{11} $a_{12} \dots \dots$ a_{1n} is called row of the matrix and the elements a_{11} $a_{21} \dots \dots$ a_{m1} is called column of the matrix. The $m \times n$ is called order of the matrix.

Example: $A = \begin{bmatrix} 2 & 8 \\ 5 & 3 \end{bmatrix}$ is the matrix of order 2 by 2.

Various Types of Matrices:

(i) **Square Matrix:** A Matrix in which the number of rows is equal to the number of columns is called a square matrix.

Example: $A = \begin{bmatrix} 4 & 5 \\ -2 & 3 \end{bmatrix}_{2\times 2}$

(ii) **Diagonal matrix:** A square matrix A is said to be diagonal matrix if all its non-diagonal elements be zero.

Example: $A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}_{3\times 3}$

(iii) **Scalar Matrix:** A square matrix A is said to be scalar matrix if all its non-diagonal elements are zero and all the diagonal elements are equal is called a scalar matrix.

Example: $A = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}_{3\times 3}$

(iv) **Unit or Identity matrix:** A square matrix A is said to be unit matrix if all its non-diagonal elements are zero and all the diagonal elements are unity is called a unit matrix.

Example: $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}$

(v) Row matrix, Column matrix, Null or zero matrix, Sub matrix

(vi) **Transpose matrix:** The matrix of order $n \times m$ obtained by interchanging the rows and columns of a matrix A of order $m \times n$ is called the transposed matrix of A. It is dented by A^t

Example: $A = \begin{bmatrix} 2 & 8 \\ 5 & 3 \end{bmatrix}$ then $A^t = \begin{bmatrix} 2 & 5 \\ 8 & 3 \end{bmatrix}$

(vii) **Symmetric matrix:** A square matrix $A = [a_{ij}]$ is called symmetric provided $a_{ij} = a_{ji}$ for all values of *i* and *j*.

Example: $A = \begin{bmatrix} 1 & -3 & 5 \\ -3 & 2 & 7 \\ 5 & 7 & 3 \end{bmatrix}$

(viii) **Skew-symmetric matrix:** A square matrix $A = [a_{ij}]$ is called skew symmetric provided $a_{ij} = -a_{ji}$ for all values of i and *j*.

Example: $A = \begin{bmatrix} 0 & 1 & -3 \\ -1 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}$

(ix) **Periodic matrix:** A square matrix A is called periodic, if $A^{k+1} = A$, where k is a positive integer.

(x) **Orthogonal matrix:** A square matrix A is called an orthogonal if $AA^t = I$, where I is an identity matrix and A^t is the transposed matrix of *A*.

Example: $A = \frac{1}{3}\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$

(xi) **Idempotent matrix:** A square matrix *A* is called idempotent provided it satisfies the relation $A^2 = A$.

Example: Show that the matrix $A = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix}$ is idempotent.

(xii) **Involutory matrix:** A square matrix *A* is called involuntary provided it satisfies the relation $A^2 = I$, where *I* is the identity matrix.

Example: Show that the matrix $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & 3 \end{bmatrix}$ is involutory.

(xiii) **Nilpotent matrix:** A square matrix *A* is called nilpotent matrix of order *m*, provided it satisfies the relation $A^m = 0$ and $A^{m-1} \neq 0$, where *m* is a positive integer and *O* is the null matrix.

Example: Show that the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ -1 & -2 & -3 \end{bmatrix}$ is a nilpotent matrix of order 2.

Eigenvalue and Eigenvector

Definition: A matrix eigenvalue problem considers the vector equation

$Ax = \lambda x$ …….(1)

Here ***A*** is a given square matrix, λ an unknown scalar, and ***x*** an unknown vector. The λ 's that satisfy equation (1) are called eigenvalues of A and the corresponding nonzero ***x***'s that also satisfy equation (1) are called eigenvectors of ***A.***

Uses

Eigenvalues have a very large number of applications in diverse fields such as in engineering, geometry, physics, mathematics, biology, environment science, economics, psychology, and other areas.